Electrons are supplied energy by incident photons but are decelerated by a voltage V. Assuming that it takes minimum energy ϕ to escape the metal and that photons have an energy hf then the conservation of energy gives:

$$\frac{hc}{\lambda} = \phi + \frac{1}{2}mv^2 - eV$$

Thus the graph for f/V has equation:

$$V = \frac{\phi + \frac{1}{2}mv^2 - hf}{e}$$

if we adjust the voltage until the electrons are not emitted so that $v = 0$

$$V = \frac{\phi - hf}{e}$$

which has gradient $-h/e$ and y-intercept ϕ/e so provides an easy way to measure planck's constant and the work function of a given metal.